Fractions for Grade 5

Teach fractions with a sharing game before introducing numbers. Project name: Roti.

Sector: K-12 education, grade 5, India

Status: concept study, written and designed by Karthik Bindumon, 2026. Numbers are targets or modelled assumptions and are labelled as such.

Method: The Realisation Loop, v1 (see 00-realisation-loop.pdf).

1 Brief

Ask a grade 5 class which is bigger, one third or one quarter, and a large share choose one quarter because four is bigger than three. Whole-number reasoning applied to fractions is one of the best-documented misconceptions in mathematics education, and it survives years of worksheets. This study designs four sessions that teach fractions as a fairness dispute first and as numbers second.

2 Audience and needs analysis

Primary audience: grade 5 students in an Indian school, class size 35 to 45, mixed home languages, one teacher, no devices. What they know: sharing food fairly, and arguing about it. What they lack: a bridge from that competence to written fractions. Constraints: forty-minute periods, syllabus pressure, a teacher who has to referee 40 arguments at once.

Needs analysis method: a ten-item diagnostic (compare unit fractions, order, equivalence) before session one; two class observations; a fifteen-minute teacher interview.

3 The realisation

> **Children already run a strict fairness court at the dinner table. They know a roti cut in four gives smaller pieces than a roti cut in three. Teach fractions as sharing disputes first and numbers second.**

Contrarian core: do not introduce the symbol 1/3 until the class has already invented it. Symbols arrive in session three, after the arguments have been had and won.

4 Learning objective (target)

After four sessions, at least 80 percent of students correctly order unit fractions with denominators 2 to 8 and name one equivalent pair, on a diagnostic that uses no roti imagery.

5 Design

Who Eats More. A table game for four. Each round, one paper roti and a card that says how many people are sharing it. Fold, cut, compare one piece against another table's piece. The round ends only when the table agrees who got more. The teacher referees disputes; the disputes are the content.

Session arc. Session 1: enactive. Cut and compare, no writing. Session 2: iconic. Draw the cut before you make it; predict who gets more. Session 3: symbolic. The class names the pieces; the

teacher writes what they say. Session 4: number line. Lay the pieces end to end; the line appears.

Implicit learning. Equivalence is never taught. It shows up the day two players cut a roti differently (two of four, one of two) and nobody complains. The teacher points at it and says nothing else.

Artefacts. Paper roti templates, round cards (2 to 8 sharers), a one-page teacher referee guide, the diagnostic, the session-end drawing sheet.

6 Theory map

Source	How it is used
Bruner, enactive to iconic to symbolic	The four-session arc in that order, no faster.
Papert, constructionism	Knowledge built through a thing the child cut with their own hands.
Montessori, manipulatives	The roti carries the concept until the symbol can.
Vygotsky, ZPD	The argument at the table is the scaffold; the teacher only referees.
Lakoff and Nunez, embodied mathematics	Fractions as portions of a whole the body already knows.
Huizinga, Homo Ludens	The rule (fair share) precedes the concept.

7 Evaluation plan (Kirkpatrick)

Level	Measure	When
1 Reaction	One drawing per child: 'draw the moment you were sure'.	End of each session
2 Learning	Diagnostic, same items, no roti imagery.	Before session 1; two weeks after session 4
3 Behaviour	Transfer: number-line items the class has never seen.	Four weeks after
4 Results	Teacher interview: what changed in ordinary fraction lessons; adoption by the next grade 5 teacher.	End of term

8 Pilot plan and risks

Two classes, one teacher each, four sessions over two weeks, one control class on the textbook chapter. Risks: noise (the referee guide has a two-minute quiet rule); a table that will not agree (the teacher records the dispute and moves on; the dispute is data); the syllabus (four periods total, mapped to the existing chapter).

9 Ethics

No screens, no data on individual children beyond the diagnostic held by the teacher. Rotis respect what the class already eats; substitute the local flatbread in other regions.

10 What I would need from the client

Two grade 5 classes, four periods each, a control class, paper and scissors, and thirty minutes with each teacher before and after.